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The Editor
Classical and Quantum Gravity

Dear Editor,

I am writing to submit the enclosed manuscript, *The Other Half of Gravity: Torsion and the Two-Chamber Structure of the Tetrad Frame*, for consideration as a Letter in Classical and Quantum Gravity.

This letter is a companion to the manuscript *The Coordinate-Space Geometry of the Tetrad Frame* (CQG-115428), currently under review at your journal. That companion letter proved that the spatial sector of the tetrad equation produces two bounded tetrahedral regions related by spatial parity: TetA (the identity component) and TetB (the parity component). The present letter addresses the question of how this two-chamber structure manifests at the level of the affine connection.

The letter contains two formal results:

1. **Lemma (Section 2).** A torsion-free connection cannot distinguish TetA from TetB. The proof follows in three steps from the bilinearity of the metric in the frame field: under the spatial parity that exchanges TetA and TetB, both frame factors acquire a sign flip, leaving the metric invariant; the Christoffel symbols, being uniquely determined by the metric, inherit this invariance. The torsion sector is therefore the necessary carrier of the two-chamber distinction at the connection level.
2. **Theorem (Section 5).** Under the closure relation $D = K - 1$, the standard torsion count $K \times K(K-1)/2$ equals the product $D \times 2K$ if and only if $K = 4$. This links the antisymmetric connection count to the coupling range of a polynomial that organises every standard count of the tetrad formalism.

A supplementary note derives the TetB field fraction $1/(x+1)$ from the exchange symmetry of the two chambers combined with the rational-bijection classification on the projective line. The dark-to-baryon mass ratio that appears in the cold dark matter framework is shown in Section 4 to be an algebraic identity of this partition, following from four lines of algebra with no halo model and no free parameters.

The manuscript does not propose a modification of general relativity. It examines the geometric consequences of relaxing the torsion-free specialisation within the standard tetrad formalism. All claims are either proved locally, derived from the companion geometry (CQG-115428), or cited from standard references (Misner, Thorne and Wheeler 1973; Hehl *et al.* 1976; Cartan 1922).

Because the present letter depends directly on the two-chamber geometry established in CQG-115428, I would be grateful if the same referee could be considered for both manuscripts, should that be practical. This would avoid the need to re-evaluate the companion result within the present

review.

The manuscript has not been submitted elsewhere and is not under consideration at any other journal.

Yours sincerely,

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